VOL. I · NO. 7 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

MONDAY, NOVEMBER 2, 2026 ALL TIMES EASTERN STANDARD TIME (EST)

THE WEEK IN LAUNCH

India's NVS-03 and the 700th Falcon 9 close out October

Two orbital launches flew worldwide since Thursday's issue. The Indian Space Research Organisation (ISRO) put the NVS-03 navigation satellite into orbit on a Geosynchronous Satellite Launch Vehicle (GSLV) Friday afternoon, and SpaceX launched Starlink 15-18 from Vandenberg just after midnight Saturday, the 700th Falcon 9 flight. That brings 2026 to 253 orbital launch attempts, 246 of them successful, according to GCAT, and October ended with every launch a success.

Nine more launches have dates in the coming week, starting with Rocket Lab's 99th Electron at 11:41 a.m. Monday. The U.S. Space Force mission USSF-432 has moved to Tuesday night in Launch Library 2. On the International Space Station, Thursday's spacewalk installed a new roll-out solar array, and the final power-upgrade spacewalk is set for November 5.

I · RECENT LAUNCHES & RETURNS

GSLV puts NVS-03 in orbit; Falcon 9 flies for the 700th time

ISRO's fourth launch of the year and SpaceX's 123rd were both successful.

GSLV: NVS-03

"LIFT-OFF! GSLV-F18 carrying NVS-03 lifts off from Satish Dhawan Space Centre, SHAR, Sriharikota," ISRO posted at 2:21 p.m. EDT Friday (18:21 UTC), 11:51 p.m. India time. Thirty-two minutes later it added: "Mission Success! GSLV-F18 has successfully placed NVS-03 in the intended orbit." The rocket flew from the Second Launch Pad (NASASpaceflight).

NVS-03, also designated IRNSS-1L, is a replacement satellite for India's regional navigation system, NavIC (Launch Library 2). GCAT records an initial transfer orbit of 178 by 37,896 kilometers, inclined 20.7 degrees. It was India's fourth orbital launch of 2026; the first, a Polar Satellite Launch Vehicle on January 12, failed.



A Falcon 9 lifts off from Space Launch Complex 4E at Vandenberg Space Force Base with NASA's SWOT satellite on Dec. 16, 2022. Starlink 15-18 flew from the same pad on Saturday.
PHOTO: NASA/KEEGAN BARBER · PUBLIC DOMAIN

Falcon 9: Starlink 15-18

"LAUNCH! SpaceX Falcon 9 B1093-18 launches Starlink Group 15-18 from SLC-4E, Vandenberg," NASASpaceflight posted just after midnight. Liftoff was at 12:04 a.m. EDT Saturday (04:04 UTC), 9:04 p.m. Friday local time (Spaceflight Now); Thursday's issue had listed the 11:48 p.m. start of the FAA airspace window. Booster B1093 was on its 18th flight, and the rocket headed for an orbit inclined 70 degrees (GCAT).

The booster landed on the droneship Of Course I Still Love You, the 675th Falcon booster landing, and SpaceX confirmed deployment of the 27 satellites at 1:08 a.m. It was the 700th Falcon 9 flight and SpaceX's 11th and final orbital launch of October (Spaceflight Now). No other orbital launches took place worldwide between Thursday's issue and Monday morning (Launch Library 2).

II · LAUNCH SCHEDULE

Electron's 99th launch opens a week of nine

Every orbital launch with a date between now and November 30. Times are EST with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Mon Nov 2, 11:41 a.m. 16:41 UTC	Electron	Rocket Lab	"The Sea God Surveys": one iQPS radar imaging satellite, 575 km, 42°	Launch Complex 1, Mahia, New Zealand	LEO	**CONFIRMED**	Rocket Lab; Launch Library 2
Mon Nov 2, 7:10 p.m. 00:10 UTC Nov 3	Kinetica 2	CAS Space	Listed as a Spacesail (G60) polar group; payload not confirmed	Launch Area 140, Jiuquan, China	Polar	**TENTATIVE**	Launch Library 2
Tue Nov 3, 8:28 p.m. 01:28 UTC Nov 4	Falcon 9	SpaceX	USSF-432, a classified U.S. Space Force mission (FAA: "R-4")	SLC-4E, Vandenberg, Calif.	LEO	**TENTATIVE**	Launch Library 2; FAA
Wed Nov 4, 12:00 a.m. 05:00 UTC	Mir (name provisional)	Agency for Defense Development, South Korea	Demonstration flight, first full orbital launch of the military small-satellite rocket	Offshore launch platform	LEO	**TENTATIVE**	Launch Library 2
Thu Nov 5, 11:00 a.m. 16:00 UTC	Long March 2C	CASC	Payload not announced	Launch Complex 3, Xichang, China	—	**TENTATIVE**	Launch Library 2
Thu Nov 5, 4:15 p.m. 21:15 UTC	Long March 12	CASC	Nine SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-2, Wenchang, China	LEO	**TENTATIVE**	Launch Library 2
Thu Nov 5, 9:21 p.m. 02:21 UTC Nov 6	Falcon 9	SpaceX	Starlink 15-21: Starlink satellites	SLC-4E, Vandenberg, Calif.	LEO	**TENTATIVE**	FAA; Launch Library 2
Fri Nov 6, 6:35 p.m. 23:35 UTC	Ceres-2	Galactic Energy	Payload not announced	Launch Area 95A, Jiuquan, China	—	**TENTATIVE**	Launch Library 2
Mon Nov 9, 10:03 p.m. 03:03 UTC Nov 10	Falcon 9	SpaceX	"R-5," a classified U.S. Space Force mission	SLC-4E, Vandenberg, Calif.	LEO	**TENTATIVE**	FAA; Launch Library 2
Fri Nov 13	Vulcan VC6L	United Launch Alliance	Vulcan Leo 1 (LV-01): 40 Amazon Leo broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	**CONFIRMED**	ULA; Launch Library 2
NET Sat Nov 14	Electron	Rocket Lab	"The Mountain God Rises": iQPS radar imaging satellite	Launch Complex 1, Mahia, New Zealand	LEO	**PRELIMINARY**	Launch Library 2
Fri Nov 20, 6:00 a.m. 11:00 UTC	Long March 8A	CASC	Nine SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-1, Wenchang, China	LEO	**TENTATIVE**	Launch Library 2
NET Mon Nov 23	Ariane 64	Arianespace	VA271: 36 Amazon Leo broadband satellites, 10th Ariane 6	ELA-4, Kourou, French Guiana	LEO	**PRELIMINARY**	Launch Library 2
Tue Nov 24, 12:03 a.m. 05:03 UTC	Soyuz-2.1b	Roscosmos	Progress MS-36 cargo ship to the International Space Station	Site 31/6, Baikonur, Kazakhstan	LEO	**TENTATIVE**	Launch Library 2
Tue Nov 24, 2:00 a.m. 07:00 UTC	Long March 12	CASC	Nine SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-2, Wenchang, China	LEO	**TENTATIVE**	Launch Library 2
NET Wed Nov 25	Long March 2F	CASC	Shenzhou 24 crew to the Tiangong station	Launch Area 91, Jiuquan, China	LEO	**PRELIMINARY**	Launch Library 2
Thu Nov 26, 3:00 p.m. 20:00 UTC	Spectrum	Isar Aerospace	Third test flight, carrying several satellites including R-Space's AT-Astra	Andøya Spaceport, Norway	SSO	**TENTATIVE**	Launch Library 2
Sun Nov 29, 11:00 a.m. 16:00 UTC	Falcon 9	SpaceX	Northrop Grumman's Cygnus NG-25 cargo ship to the space station	SLC-40, Cape Canaveral, Fla.	LEO	**TENTATIVE**	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; ELA-4, Ariane Launch Area 4; LEO, low Earth orbit; SSO, Sun-synchronous orbit; CASC, China Aerospace Science and Technology Corp.; iQPS, Institute for Q-shu Pioneers of Space, a Japanese radar imaging company.

Date changes

Electron "The Sea God Surveys." Monday at 3:45 a.m.; now Monday at 11:41 a.m., the T-0 Rocket Lab set Monday morning. No reason given.

USSF-432. Monday at 8:42 p.m.; now Tuesday at 8:28 p.m., the former backup time (Launch Library 2). Monday's FAA plan still lists the Monday window as primary (FAA). No reason given.

Starlink 15-21. Thursday at 11:59 p.m.; now Thursday at 9:21 p.m., with a backup at 9:06 p.m. Friday (FAA). No reason given.

Ceres-2. Wednesday, November 4, at 4 p.m.; now Friday, November 6, at 6:35 p.m. (Launch Library 2). No reason given.

Long March 12. Thursday at 3 p.m.; now Thursday at 4:15 p.m. (Launch Library 2). No reason given.

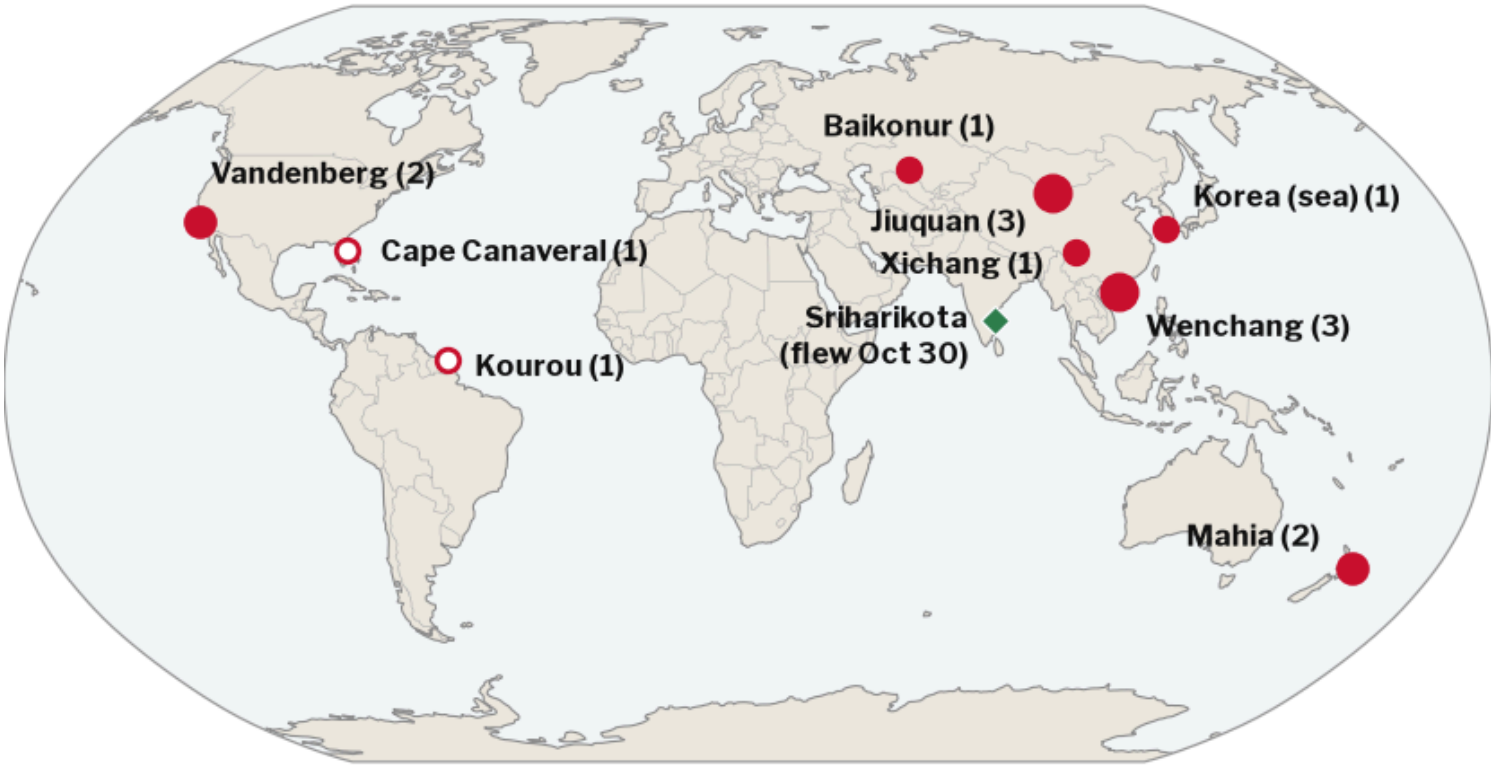
Electron "The Mountain God Rises." NET November 10; now NET November 14 (Launch Library 2). No reason given.

New to the schedule. A Long March 2C from Xichang on Thursday, a Long March 8A and a second Long March 12 from Wenchang, Isar Aerospace's third Spectrum, and a November 29 date for Cygnus NG-25 (Launch Library 2).

Also listed for November without a specific date in Launch Library 2: Starship Flight 15 from Starbase; NROL-195 and OneWeb 21 on Falcon 9s from Vandenberg; and ISRO's first uncrewed Gaganyaan test flight on a Launch Vehicle Mark 3 and a PSLV-XL with the TDS-01 technology demonstration satellite.

QUICK HIT

Electron No. 99. "Electron is vertical on Pad A at Launch Complex 1 to begin fueling," Rocket Lab posted at 7:55 a.m. Monday. It is the company's 10th dedicated mission for iQPS (Rocket Lab).



Launch sites with orbital launches dated in Launch Library 2 through November 25; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary in Launch Library 2; for Cape Canaveral's Vulcan, ULA has announced November 13. Green: launched since Thursday. Vandenberg also launched Saturday.

MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Airspace windows for three more Vandenberg Falcons

The FAA's operations plan lists the times U.S. airspace is set aside; it gives no hazard-area coordinates.

How they fly

Saturday's Falcon 9 reached an orbit inclined 70 degrees, and Friday's GSLV a transfer orbit inclined 20.7 degrees (GCAT). Monday's Electron is aiming for a 575-kilometer orbit inclined 42 degrees (Rocket Lab). All three Falcon 9s in the FAA's plan launch from Space Launch Complex 4E at Vandenberg.

Airspace windows

The FAA Air Traffic Control System Command Center's plan for Monday lists these launch windows, converted from UTC (FAA advisory):

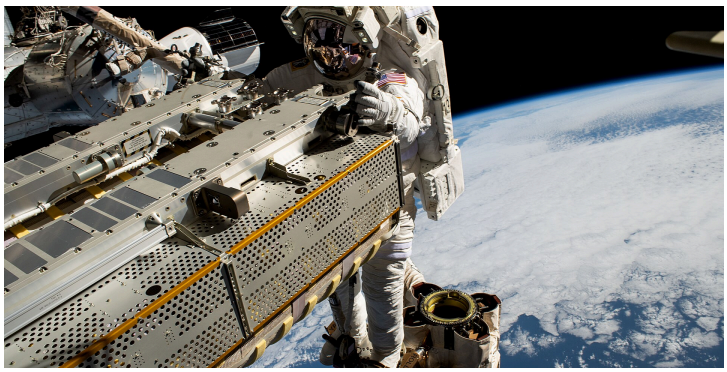
FLIGHT	PRIMARY WINDOW	BACKUP
"R-4" (USSF-432)	**Mon Nov 2, 8:42 p.m.–Tue 1:25 a.m.**	**Tue Nov 3, 8:28 p.m.–Wed 1:11 a.m.**
Starlink 15-21	Thu Nov 5, 9:21 p.m.–Fri 2:04 a.m.	Fri Nov 6, 9:06 p.m.–Sat 1:49 a.m.
"R-5"	Mon Nov 9, 10:03 p.m.–Tue 2:46 a.m.	Tue Nov 10, 9:49 p.m.–Wed 2:32 a.m.

IV · REENTRIES & ISS TRAFFIC

Nine objects come down, including a 1969 Agena fragment

CelesTrak lists three Starlinks, an Electron upper stage and the Rassvet-3 satellite that was still up on Thursday among objects decayed since October 28.

Station traffic: one spacewalk left



NASA astronaut Woody Hoburg works on a Roll-Out Solar Array outside the International Space Station during a 2023 spacewalk.
PHOTO: NASA · PUBLIC DOMAIN

NASA astronaut Anil Menon and Canadian Space Agency astronaut Joshua Kutryk "completed a spacewalk at 3:08 p.m. EDT today after installing a roll-out solar array for power generation," NASA's space station account posted Thursday. "They will conduct a final power upgrade spacewalk on Thursday, Nov. 5." The spacewalk was underway by 8:13 a.m. (NASA), and the array started to unroll during the afternoon (Spaceflight Now). The Expedition 75 crew spent Friday on space-crop and technology research (NASA).

The next visitors are Progress MS-36 at 12:03 a.m. November 24 from Baikonur, China's Shenzhou 24 to Tiangong NET November 25, with the first astronaut from Pakistan expected aboard, and Cygnus NG-25, now listed for 11 a.m. November 29 (Launch Library 2).

Confirmed reentries

CelesTrak's satellite catalog lists these objects as decayed since Thursday's issue. Dates are UTC.

OBJECT (CATALOG NO.)	LAUNCHED	DECAYED	LAST ORBIT, KM
Starlink-30233 (57302)	Jul 10, 2023	Oct 28	213 × 223
Flock 4G-20 (62636)	Jan 14, 2025	Oct 28	161 × 175
Starlink-1309 (45414)	Mar 18, 2020	Oct 29	144 × 149
Rassvet-3 17 (100083)	**Jul 19, 2026**	**Oct 29**	**149 × 159**
IOD-1 (63212)	Mar 15, 2025	Oct 30	186 × 196
Electron upper stage (101147)	Oct 23, 2026	Oct 30	126 × 182
Thor-Agena D debris (4133)	Sep 30, 1969	Oct 31	201 × 209
Cosmos 1275 debris (13809)	Jun 4, 1981	Oct 31	260 × 276
Starlink-1283 (45389)	Mar 18, 2020	Nov 1	147 × 150

Rassvet-3 17, which Thursday's issue listed as still in orbit at 187 by 192 kilometers, decayed on October 29 (CelesTrak). The Electron upper stage came from Rocket Lab's October 23 launch for HawkEye 360 and lasted a week in orbit. IOD-1 is Spanish-owned and Flock 4G-20 is a Planet imaging satellite; both were launched from Vandenberg (CelesTrak).

Rocket Lab's 100th Electron leaves the factory

Neutron's upper stage reached its Virginia pad, Blue Origin rebuilt its New Glenn tower, and both Flight 15 Starship vehicles finished their static fires.

Rocket Lab

"Triple digits unlocked," Rocket Lab posted Thursday. "The 100th Electron has left the factory, headed for launch." Monday's mission is its 99th, "our last double-digit launch," the company said Sunday. "Neutron's Stage 2 and Thrust Module have arrived to Launch Complex 3," it posted Friday, "another major step forward for the team that brings us closer to Neutron's debut"; the hardware came by sea from Middle River, Maryland, to the Mid-Atlantic Regional Spaceport at Wallops, Virginia (Rocket Lab). Rocket Lab Robotics also unveiled a robotic arm developed with NASA that "can walk across spacecraft, use tools, and perform repairs and upgrades in orbit"; it launches on NASA's Fly Foundational Robots mission in late 2027 (post).

Blue Origin

Blue Origin is targeting a return to flight for New Glenn by the end of 2026 after the May hot-fire anomaly at Launch Complex 36A, NASASpaceflight reported. Chief executive Dave Limp confirmed on October 24 that all nine modules of the pad's Vehicle Access Tower are repaired and restacked, and crews made the first major concrete pour for a second pad, LC-36B, using a convoy of 450 trucks. Hull welding of the Blue Moon Mark 2 test article that will fly on Artemis III is expected to finish by January 2027. The company also introduced ATLAS, "the first U.S. Department of Labor registered apprenticeship built exclusively for space manufacturing" (Blue Origin).

Under NASA's revised Artemis III plan, the Mark 2 test article will launch on New Glenn to low Earth orbit ahead of Orion, and the crew will spend two days docked to it, checking life support, avionics and manual controls. NASA will choose between Blue Moon and SpaceX's Starship for the Artemis IV landing, targeted for 2028, based on how both perform on Artemis III (NASASpaceflight).



Vulcan lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station on its second certification flight on Oct. 4, 2024.

U.S. SPACE FORCE PHOTO BY AIRMAN 1ST CLASS COLLIN WESSON · PUBLIC DOMAIN

United Launch Alliance

Vulcan's return to flight with 40 Amazon Leo satellites remains "planned for Friday, Nov. 13, pending range approval" (ULA). It will fly with redesigned solid rocket booster nozzles (Spaceflight Now).

Starship

Booster 22 fired all 33 Raptor engines for a full-duration 15-second static fire on October 25, and Ship 42 completed a single-engine burn simulating a deorbit on October 26 and a 60-second, six-engine static fire on October 27, finishing engine testing for Flight 15, NASASpaceflight reported. Ship 43, for Flight 16, completed cryogenic proof tests and practice catches with the Pad 2 tower arms.

"We expect Starship to fly weekly at some point in the first half next year and then second half next year, twice weekly or maybe even more frequently by the end of next year," SpaceX founder Elon Musk said on the company's third-quarter earnings call, adding that SpaceX plans to try to catch the Starship upper stage on Flight 15, expected in mid-November (Spaceflight Now).

Falcon slows down

SpaceX's 11 launches in October compare with 13 in each of January, June, July and August. The October 27 Starlink 15-16 launch was the first time since July 2025 that SpaceX's pace fell below 150 orbital launches in a 365-day period, down from a peak of 171 in December 2025, and SpaceX stopped flying Starlink missions from Florida on Falcon 9 in August, Spaceflight Now reported. Of Course I Still Love You, the Pacific droneship that caught B1093 on Saturday, has now hosted 231 landings.

QUICK HIT

Leo 2 is ready. A second Vulcan is already assembled for the Vulcan Leo 2 mission (Spaceflight Now).

Comment deadlines

Part 450. Comments on five proposed changes to the commercial launch and reentry licensing rule are due Wednesday, November 4 (docket FAA-2026-9935).

FCC space licensing. Comments on the Federal Communications Commission's Space Modernization for the 21st Century further notice (SB Docket No. 25-306) are due November 25 and replies December 28 (Federal Register).

Starbase. Scoping comments on up to 456 Starship launches a year at Starbase are due November 30, with meetings November 17 to 19 (Federal Register).

VI · BY THE NUMBERS

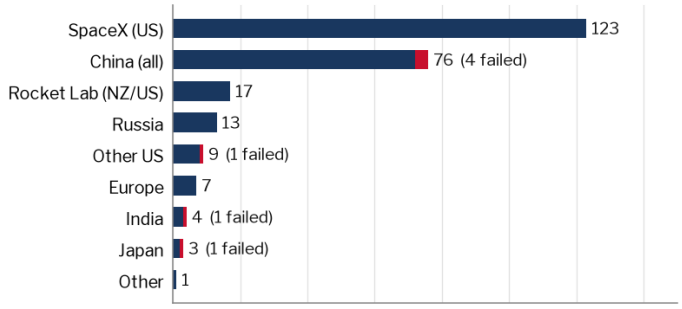
253 orbital launch attempts so far in 2026

Every one of October's 21 launches succeeded, and 11 of them were SpaceX Falcons.

253	246	123	76	21
orbital launch attempts worldwide, Jan 1–Oct 31	reached orbit	by SpaceX, all successful	by China, four failures	launches in October, all successful

The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 253 orbital launch attempts through Saturday's Starlink 15-18, with 246 successes. SpaceX has flown 123, all successful, followed by China with 76, Rocket Lab with 17 and Russia with 13; other U.S. providers have flown nine, Europe seven, India four and Japan three. GCAT counts Japan's October 19 launch of the Martian Moons eXploration (MMX) probe as a deep-space launch, outside these totals.



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure. "Other" is South Korea.

CHART: THE LAUNCH REPORT · GCAT (JONATHAN MCDOWELL) THROUGH OCT 31, 04:04 UTC

The failures

According to GCAT, seven launches have failed this year: India's PSLV on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure. Galactic Energy's Ceres-2 is now listed to fly again on November 6 (Launch Library 2).

October in full

Eleven of October's 21 launches were SpaceX Falcons, the last of them Saturday's Starlink 15-18 (GCAT). Vandenberg hosted seven of them, more than any other site, and Cape Canaveral and Kennedy four between them. China flew six, and South Korea's Nuri, Japan's H3, Rocket Lab's Electron and India's GSLV account for the other four.

The week ahead

Nine orbital launches have specific dates between Monday and November 9 (Launch Library 2; FAA): three SpaceX Falcon 9s from Vandenberg, four Chinese rockets, Rocket Lab's 99th Electron and South Korea's offshore demonstration flight.

Takeaways

1. ISRO's GSLV put NVS-03 in orbit Friday, and Starlink 15-18 was the 700th Falcon 9 flight.
2. All 21 of October's launches succeeded; 2026 stands at 253 orbital attempts.
3. Rocket Lab's 99th Electron is set for 11:41 a.m. Monday, and its 100th has left the factory.
4. USSF-432 has moved to Tuesday at 8:28 p.m. in Launch Library 2.
5. The last space station power-upgrade spacewalk is set for November 5.

KEY DATES AHEAD (EST)

Mon Nov 2	Electron iQPS, 11:41 a.m.; Kinetica 2, 7:10 p.m.
Tue Nov 3	USSF-432, 8:28 p.m.
Wed Nov 4	South Korean demo flight; Part 450 comments due
Thu Nov 5	Space station spacewalk; Long March 2C and 12; Starlink 15-21, 9:21 p.m.
Fri Nov 6	Ceres-2, 6:35 p.m.
Fri Nov 13	Vulcan Leo 1
Wed Nov 25	Shenzhou 24 NET; FCC comments due

SOURCES

Schedules: Launch Library 2 (The Space Devs), Spaceflight Now, NASASpaceflight, FAA operations plan advisory. Agencies: NASA, ISRO, FAA, FCC, Federal Register. Tracking: CelesTrak, GCAT (Jonathan McDowell). X posts: @isro, @RocketLab, @SpaceflightNow, @NASASpaceflight, @SpaceX, @NASA, @Space_Station, @blueorigin, @ulalaunch. Photos: NASA and U.S. Space Force (public domain). Maps: Natural Earth (public domain).

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